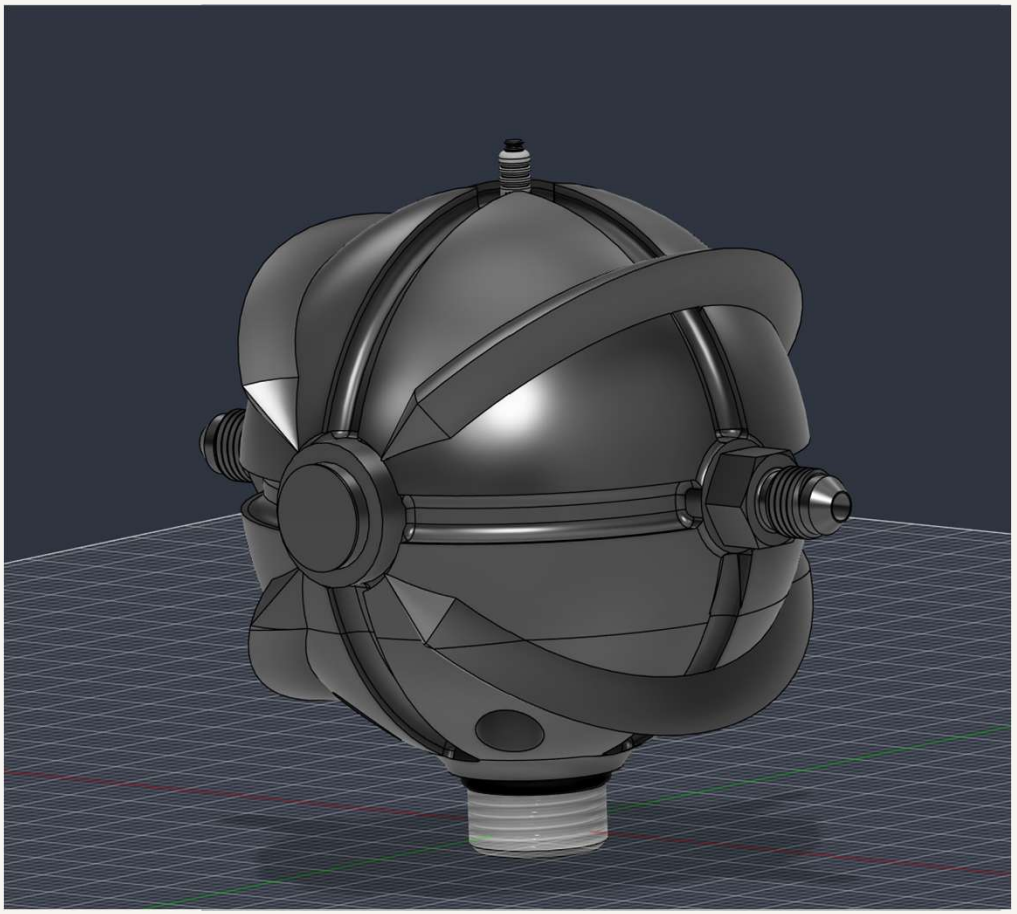
Tushar Bhat

Engineering Projects

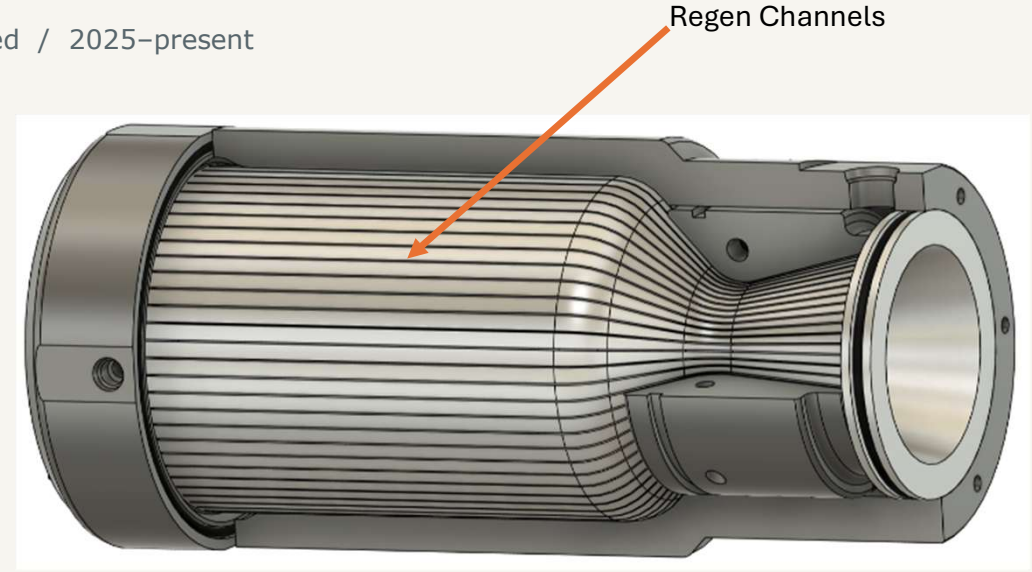


Aerospace Engineering / Purdue University
Graduation: Spring 2028

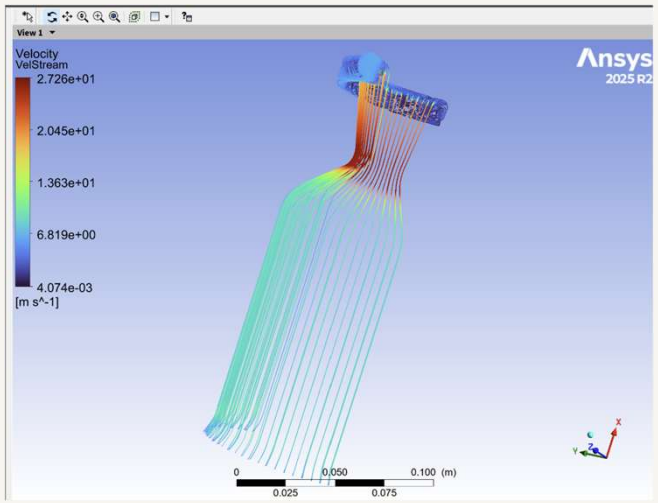
Liquid Rocket Engine Analysis

Purdue Undergraduate Rocket Propulsion Labs / Testbed / 2025–present

On the six-person simulation team, I ran thermal and fluid analysis for our regeneratively cooled RP1-LOx engine. Most of my focus was on the regen channels as they will define if the engine can handle the temperatures and pressures without exploding.



Completed CAD of the Regen Engine



Completed CFD Analysis of our regen channels

* This project is still in the early stages and will be updated a lot over the next couple of weeks!

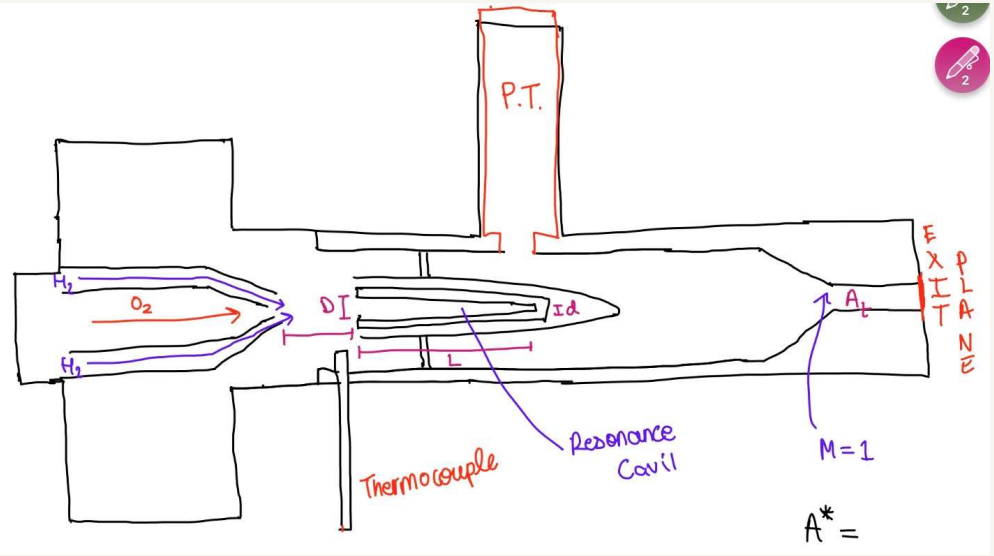
Club’s First Resonance Igniter

Purdue Undergraduate Rocket Propulsion Labs / Concept development

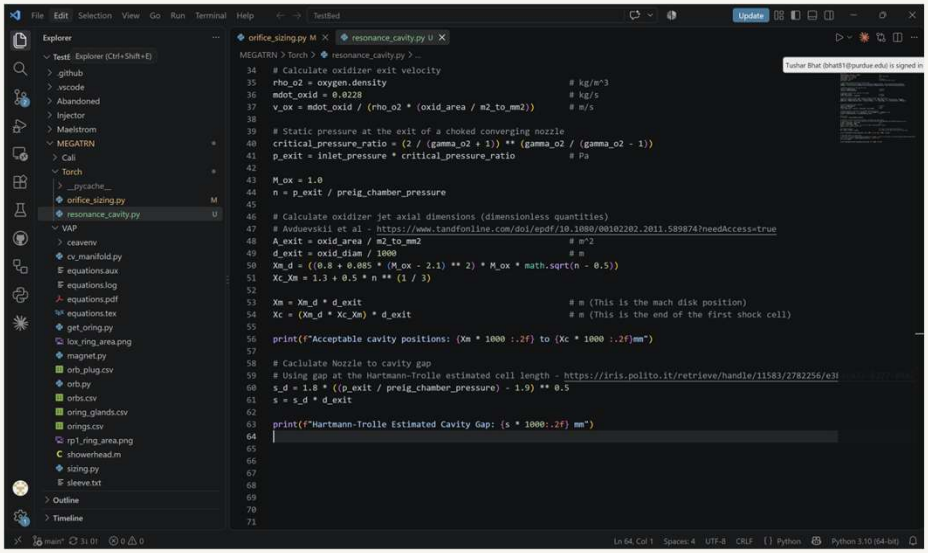
Started design to reduce the load on test infrastructure. Resonance igniters don’t require much electrical work like traditional spark plug igniters

Our work so far covers the requirements, propellant choice and preliminary sizing, utilizing extensive research on past designs.

Our first concept sketch

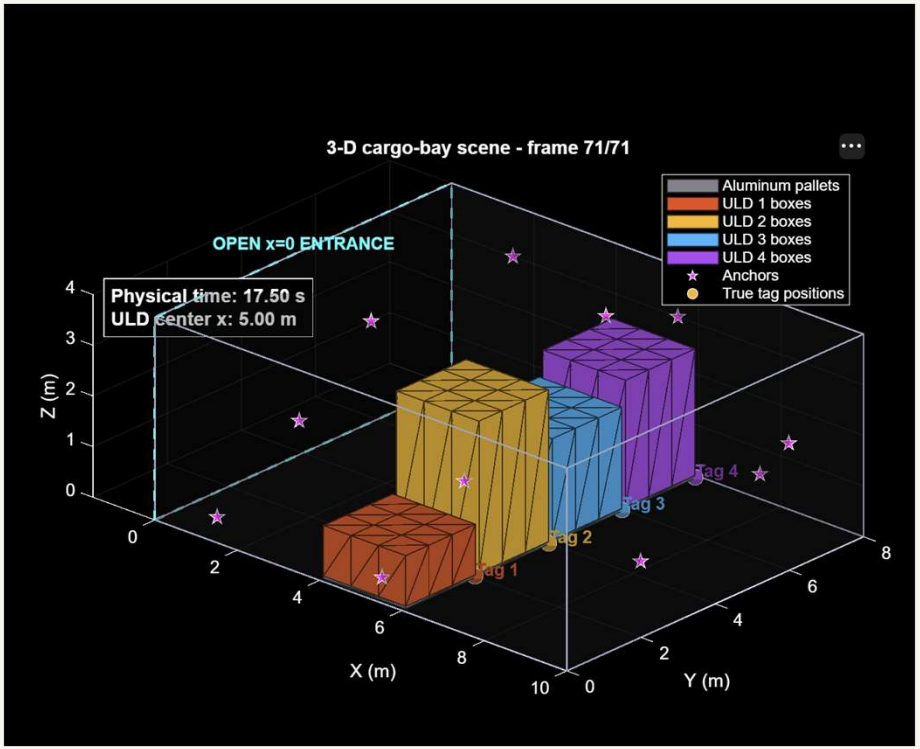


Some code from our resonance cavity sizing



Locating Objects with Bluetooth Low Energy

HCLTech / Systems R&D intern / Summer 2026



Very early, simple design of our model. Unfortunately can’t share later versions ;(

I researched and built a proof of concept MATLAB model to estimate location from Bluetooth signal strength in a metal cargo environment. Ray tracing let me test the approach without access to a physical aircraft interior.

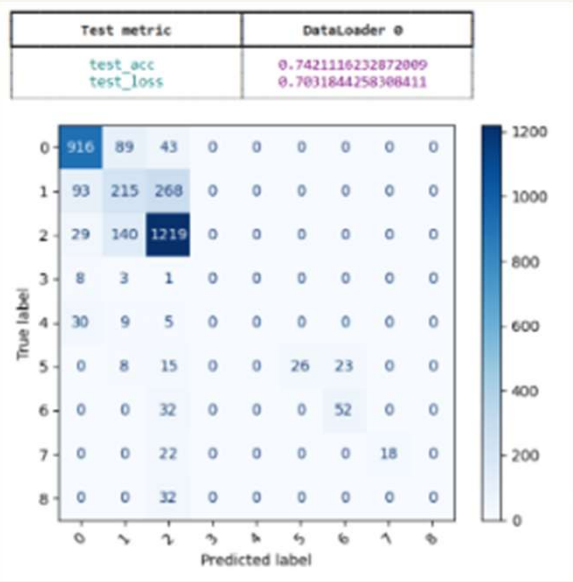
Concept to working simulation, with the system layout and tracking approach developed alongside a senior engineering manager.

Classifying interference in radio signals

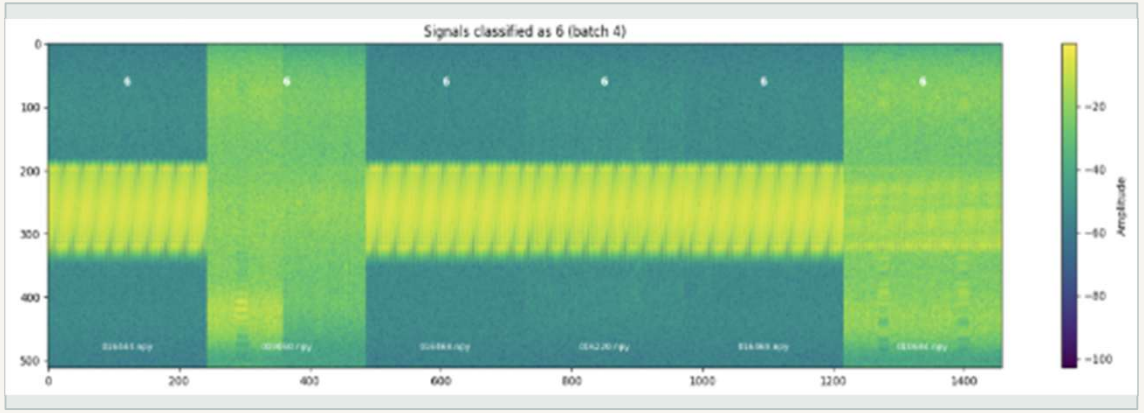
Purdue Data Mine + The Aerospace Corporation / 2025–26

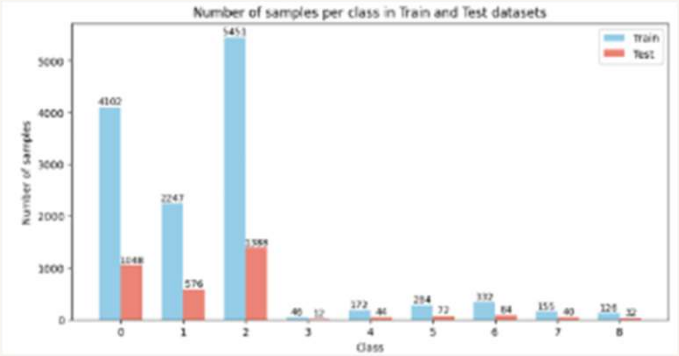
Built a signal classification model and worked on rare classes and noisy labels. My model reached accuracy of around 74%. My team's 11 model ensemble reached 76.91% test accuracy.

Test metric	DataLoader 0
test_acc	0.7421116232872009
test_loss	0.7031844258308411



How a spectrogram was visualized given the data we had





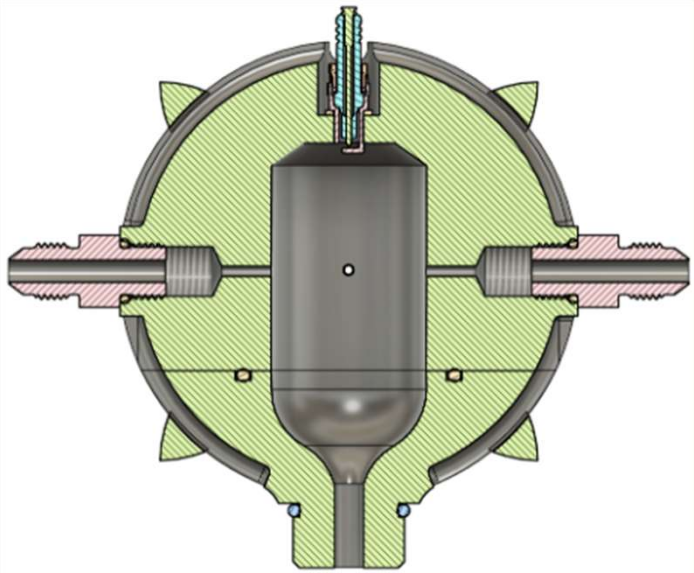
Imbalanced Data made the classification really hard and skewed towards the former classes

Design and Build a Torch Igniter

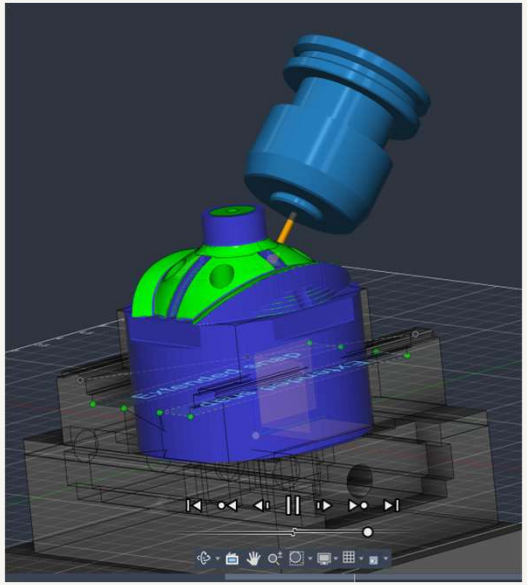
Independent project / 2025–present

I designed and CNC machined a methane and oxygen igniter, then compared a combustion simulation with an ideal reference I got during preliminary design

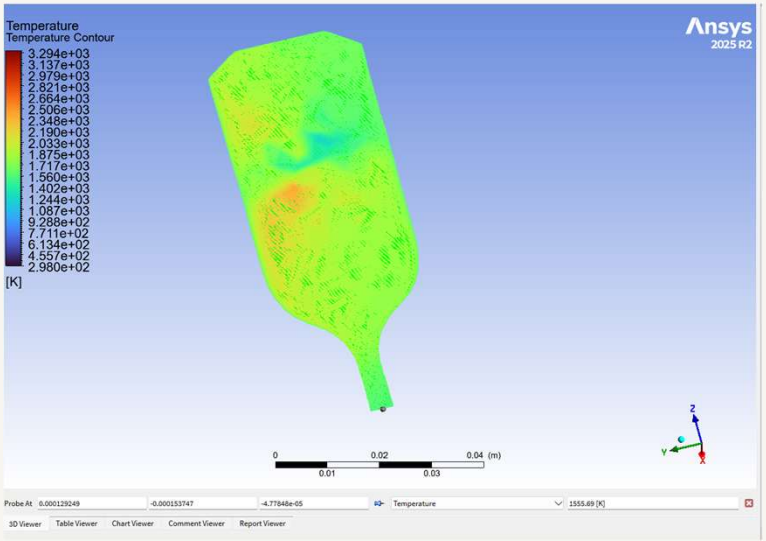
The simulation reached 84% efficiency and an estimated 1,600 K exit temperature.



CAD and Section Analysis of Igniter



CAM toolpath



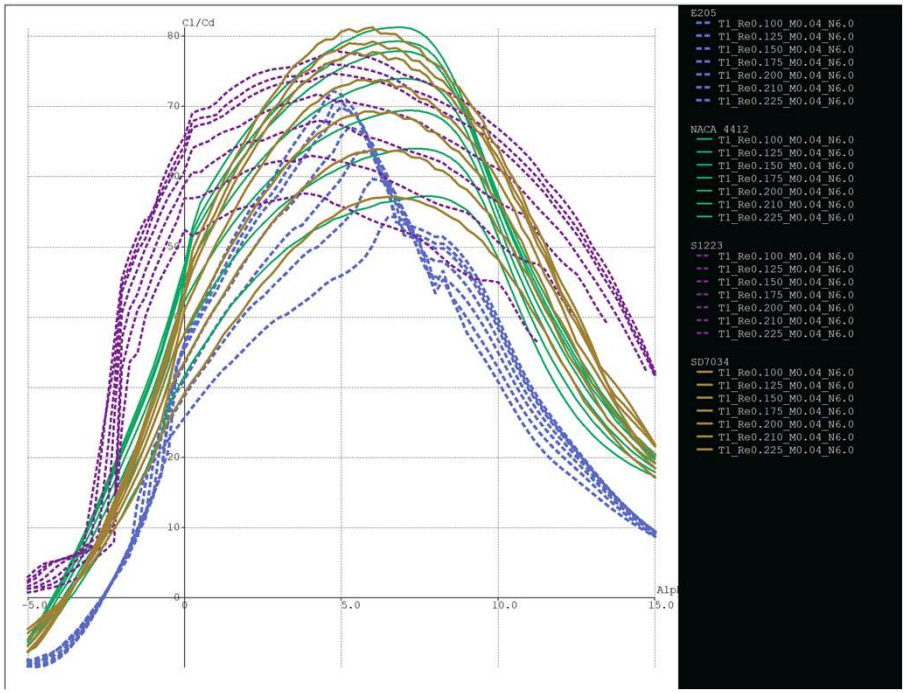
Temperature of the torch fluid

An Autonomous UAV

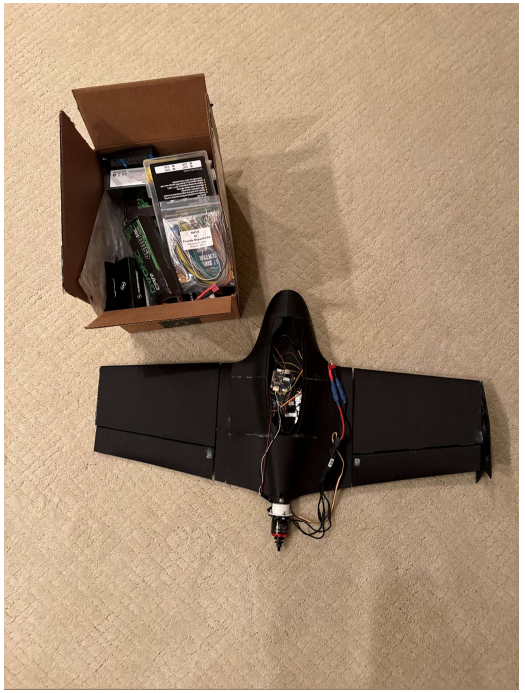
Personal project / 2025

I designed the airframe and bench tested its electronics.

The design uses a 0.8 m wingspan with a 30 mph target. I used XFLR5 and NX for the aircraft, then integrated the flight controller, Raspberry Pi, GPS, motors and servos.



One of the graphs from XFLR5 to analyze wing geometries



Cleaned build after first flight crash

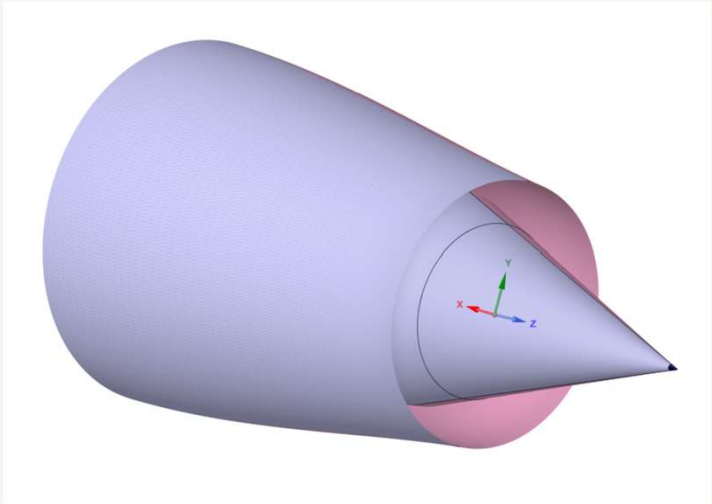
A Mach 3.2 ramjet inlet

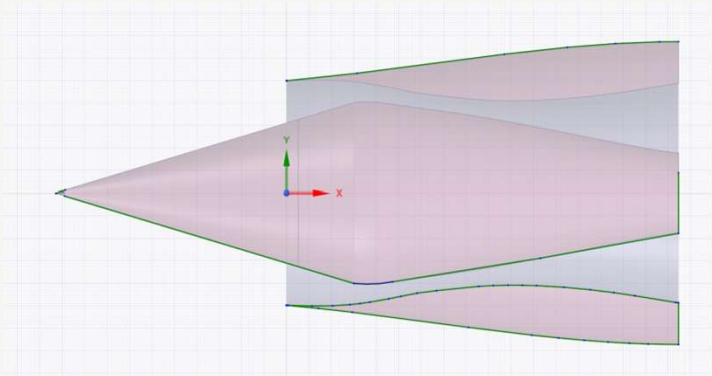
Collaborative design project / Inlet design lead

My current design concentration is the inlet

We are utilizing NASA's SUPIN software to compare inlet shapes and pressure recovery for a test focused axisymmetric ramjet. The design is still evolving.

Had to conduct a lot of research and trade studies to find the optimal design parameters such as cone half angles, cone counts, subsonic diffuser lengths, etc.





Our current inlet STL file generated with the help of SUPIN given our parameters and constraints. Needs to be converted to CAD